

Predictive Analytics Using Historical Stock Price Data

1. Introduction

Predictive analytics is a technique used to predict future outcomes by analyzing historical data. It uses statistical methods and machine learning algorithms to identify patterns and trends from past data.

In this project, historical stock price data is used to predict future stock closing prices. Stock price prediction helps understand market trends and supports decision-making.

2. Objective

The main objective of this project is to build a predictive model using historical stock price data and forecast future stock closing prices.

The objectives of this project are:

- To collect and analyze historical stock price data
- To preprocess the dataset
- To train a machine learning model
- To predict future stock closing prices
- To evaluate the model performance
- To visualize actual, predicted, and forecasted stock prices

3. Dataset Description

The dataset used in this project contains historical stock price records for 90 days.

The dataset includes the following columns:

- Date: Represents the date of the stock record
- Open: Opening price of the stock
- High: Highest price of the stock on that day
- Low: Lowest price of the stock on that day
- Close: Closing price of the stock
- Volume: Number of shares traded

In this project, the Close column is used as the target variable because closing price is commonly used for stock price prediction.

4. Tools and Technologies Used

The following tools and technologies were used in this project:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Linear Regression Algorithm

5. Data Preprocessing

Data preprocessing is an important step in machine learning. It helps prepare the dataset for model training.

The preprocessing steps performed in this project are:

1. Loaded the stock price dataset using Pandas
2. Checked dataset information
3. Checked missing values
4. Converted the Date column into datetime format
5. Created a new numerical column called Day
6. Used the Day column as input for model training
7. Used the Close column as output for prediction

Since machine learning models cannot directly understand date values, the Date column was converted into a numerical Day column.

6. Model Used

The machine learning model used in this project is Linear Regression.

Linear Regression is a supervised machine learning algorithm used to predict continuous values. It finds the relationship between an independent variable and a dependent variable.

In this project:

- Independent Variable: Day
- Dependent Variable: Closing Price

The model learns the relationship between days and stock closing prices and predicts future closing prices.

7. Methodology

The project was completed using the following steps:

1. Imported required Python libraries

2. Loaded the stock price dataset
3. Converted the Date column into datetime format
4. Created a Day column for numerical representation
5. Visualized historical stock closing prices
6. Prepared input and output variables
7. Trained the Linear Regression model
8. Predicted closing prices for existing data
9. Evaluated the model using performance metrics
10. Forecasted stock closing prices for the next 7 days
11. Visualized actual, predicted, and future forecasted values
12. Saved the output files

8. Model Evaluation

The model was evaluated using the following performance metrics:

- MAE: Mean Absolute Error measures the average absolute difference between actual and predicted values.
- MSE: Mean Squared Error measures the average squared difference between actual and predicted values.
- RMSE: Root Mean Squared Error shows prediction error in the same unit as the target value.
- R2 Score: Shows how well the model explains the variation in the data.

A higher R2 Score indicates better model performance.

9. Results

The Linear Regression model successfully predicted the stock closing prices based on historical data.

The model also forecasted the stock closing prices for the next 7 days. The results were visualized using graphs showing:

- Actual closing price
- Predicted closing price
- Future forecasted closing price

The graph clearly shows the trend of stock prices over time.

10. Output Files

The following output files were generated:

- stock_price_dataset.csv: Original stock price dataset
- stock_price_with_predictions.csv: Dataset with actual and predicted closing prices
- future_stock_price_forecast.csv: Future 7-day stock price forecast
- stock_prediction_graph.png: Graph showing actual, predicted, and forecasted values

11. Advantages

The advantages of this project are:

- Helps understand future stock price trends
- Uses historical data for prediction
- Easy to implement using Python
- Provides visual representation of trends
- Helps in learning predictive analytics concepts

12. Limitations

The limitations of this project are:

- Stock prices are affected by many external factors
- Linear Regression may not capture complex market behavior
- The dataset size is small
- Predictions are only approximate
- Real-world stock prediction requires more advanced models and larger datasets

13. Conclusion

This project demonstrates how predictive analytics can be used to forecast future trends using historical stock price data. The dataset was preprocessed, visualized, and used to train a Linear Regression model.

The model predicted stock closing prices and forecasted future values for the next 7 days. This project helped understand the basic workflow of predictive analytics, including data preprocessing, model training, evaluation, forecasting, and visualization.

14. Future Scope

In the future, this project can be improved by:

- Using a larger real-time stock market dataset
- Applying advanced machine learning models
- Using algorithms like Random Forest, ARIMA, LSTM, or XGBoost
- Including additional features like market news and trading indicators
- Building a dashboard for interactive visualization

15. Viva Explanation

I completed a predictive analytics project using historical stock price data. First, I loaded and analyzed the dataset. Then I converted the Date column into datetime format and created a numerical Day column. I used the Day column as input and the Close price as output. After that, I trained a Linear Regression model to predict stock closing prices. The model was evaluated using MAE, MSE, RMSE, and R2 Score. Finally, I forecasted the stock

closing price for the next 7 days and visualized the actual, predicted, and future forecasted values using graphs.

16. Screenshots & Outputs :

First 5 rows of dataset:

	Date	Open	High	Low	Close	Volume
0	2024-01-01	1513.87	1528.76	1478.55	1512.42	55486
1	2024-01-02	1513.10	1541.16	1470.12	1498.57	189407
2	2024-01-03	1509.71	1537.26	1485.85	1520.24	136900
3	2024-01-04	1539.23	1554.75	1513.94	1544.14	176020
4	2024-01-05	1496.35	1528.37	1475.86	1502.24	152795

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 90 entries, 0 to 89

Data columns (total 6 columns):

```
#   Column  Non-Null Count  Dtype
```

```
---  ---  -
```

```
0   Date    90 non-null    object
1   Open    90 non-null    float64
2   High    90 non-null    float64
3   Low     90 non-null    float64
4   Close   90 non-null    float64
5   Volume  90 non-null    int64
```

```
dtypes: float64(4), int64(1), object(1)
```

```
memory usage: 4.3+ KB
```

```
None
```

Statistical Summary:

	Open	High	Low	Close	Volume
count	90.000000	90.000000	90.000000	90.000000	90.000000
mean	1588.347556	1616.637000	1559.706333	1587.530222	142593.911111
std	62.444816	59.724619	59.243606	59.144033	61816.720695
min	1466.430000	1512.630000	1438.830000	1478.460000	51062.000000
25%	1537.502500	1567.492500	1512.995000	1540.277500	81164.000000
50%	1585.980000	1615.080000	1555.610000	1585.690000	139855.000000
75%	1641.942500	1669.495000	1611.870000	1641.530000	200509.750000
max	1733.630000	1750.630000	1681.900000	1702.790000	248713.000000

Missing Values:

Date 0
Open 0
High 0
Low 0
Close 0
Volume 0
dtype: int64

Dataset after converting Date column:

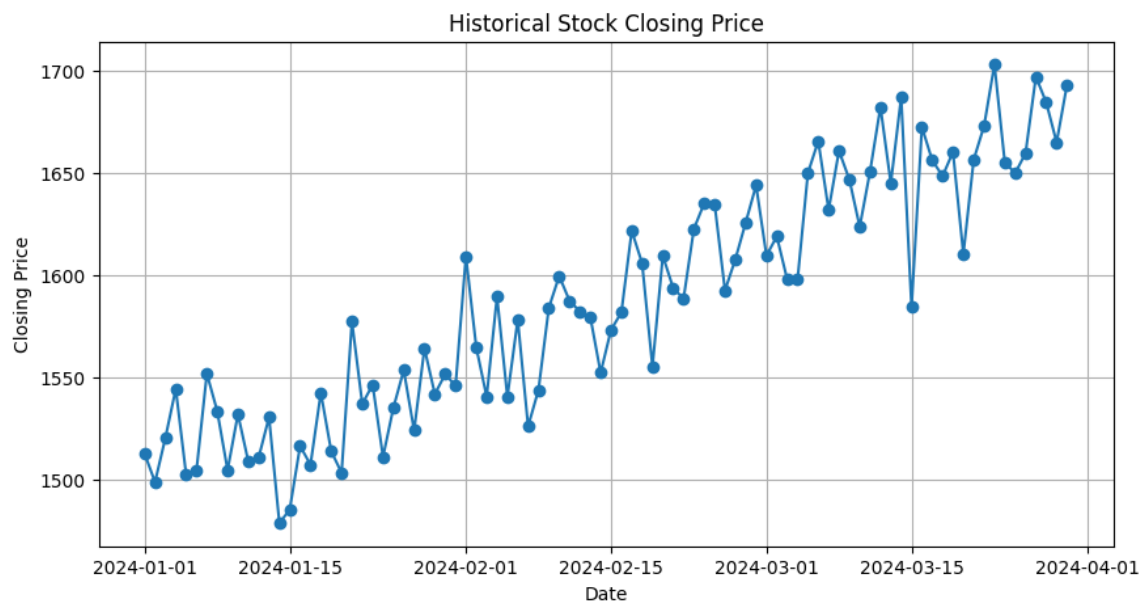
	Date	Open	High	Low	Close	Volume
0	2024-01-01	1513.87	1528.76	1478.55	1512.42	55486
1	2024-01-02	1513.10	1541.16	1470.12	1498.57	189407
2	2024-01-03	1509.71	1537.26	1485.85	1520.24	136900

3 2024-01-04 1539.23 1554.75 1513.94 1544.14 176020

4 2024-01-05 1496.35 1528.37 1475.86 1502.24 152795

Dataset with Day column:

	Date	Open	High	Low	Close	Volume	Day
0	2024-01-01	1513.87	1528.76	1478.55	1512.42	55486	1
1	2024-01-02	1513.10	1541.16	1470.12	1498.57	189407	2
2	2024-01-03	1509.71	1537.26	1485.85	1520.24	136900	3
3	2024-01-04	1539.23	1554.75	1513.94	1544.14	176020	4
4	2024-01-05	1496.35	1528.37	1475.86	1502.24	152795	5



Actual and Predicted Closing Prices:

	Date	Close	Predicted_Close
0	2024-01-01	1512.42	1494.950977
1	2024-01-02	1498.57	1497.031409
2	2024-01-03	1520.24	1499.111842
3	2024-01-04	1544.14	1501.192274

4 2024-01-05 1502.24 1503.272707

Model Evaluation Results:

Mean Absolute Error: 18.50467409023214

Mean Squared Error: 537.9759803807752

Root Mean Squared Error: 23.194309224048368

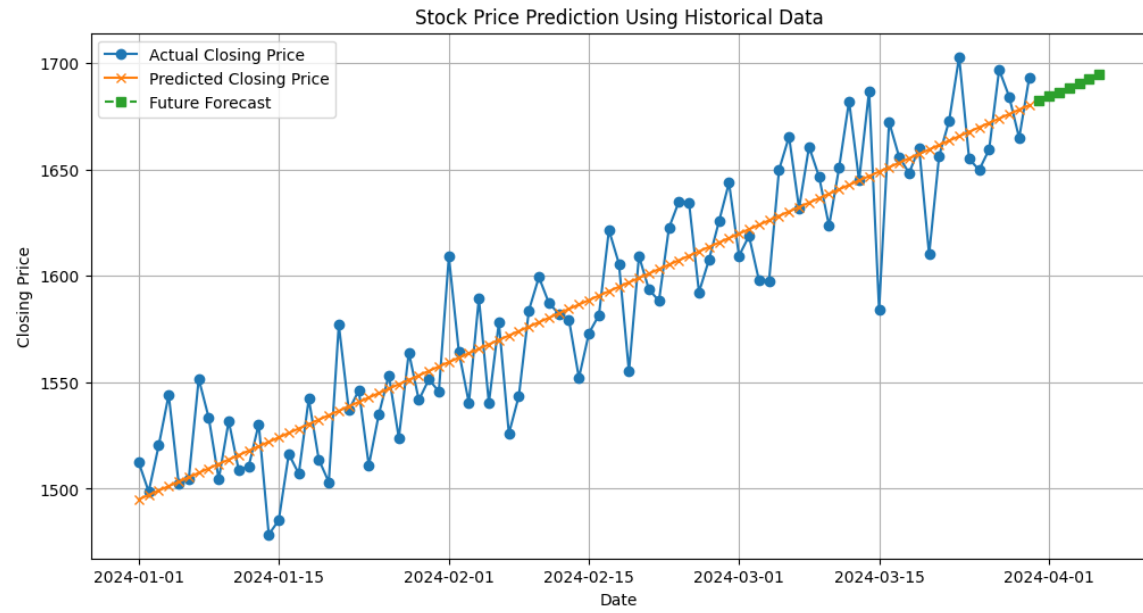
R2 Score: 0.8444773985937312

Future Stock Price Forecast:

	Date	Predicted_Close
0	2024-03-31	1682.189900
1	2024-04-01	1684.270333
2	2024-04-02	1686.350765
3	2024-04-03	1688.431198
4	2024-04-04	1690.511630
5	2024-04-05	1692.592063
6	2024-04-06	1694.672495

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
X does not have valid feature names, but LinearRegression was fitted with feature names

warnings.warn([



Note:-I Partially take help from ai to remake this Report